

FIITJEE

ALL INDIA TEST SERIES

FULL TEST – I

JEE (Main)-2024

TEST DATE: 28-12-2023

ANSWERS, HINTS & SOLUTIONS

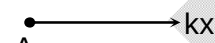
Physics

PART – A

SECTION – A

1. A

Sol. $a_1 = \frac{F}{m} - a$


A
 $ma = kx$


 $F - kx = ma_1$

2. B

Sol. $F_x = -\frac{\partial U}{\partial x} = -\frac{\partial}{\partial x}(\alpha x - \beta y) = -\alpha$

$$F_y = -\frac{\partial U}{\partial y} = -\frac{\partial}{\partial y}(\alpha x - \beta y) = +\beta$$

$$\therefore F = \sqrt{F_x^2 + F_y^2} = \sqrt{\alpha^2 + \beta^2}$$

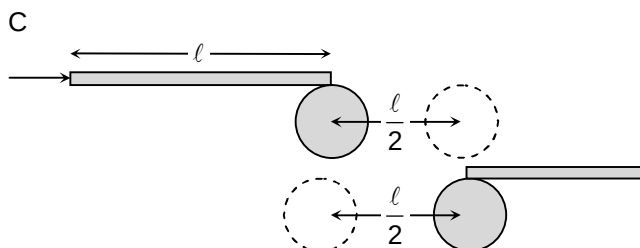
$$\therefore a = \frac{F}{m} = \frac{\sqrt{\alpha^2 + \beta^2}}{m} = \left(\frac{\alpha^2 + \beta^2}{m^2} \right)^{1/2}$$

3. D

Sol. $V = \sqrt{V_{\parallel}^2 + V_{\perp}^2} \quad e = \frac{V_2 - V_1}{U_1 - U_2}$

$$= \sqrt{(2\sqrt{2})^2 + (6\sqrt{2})^2} \quad \ell = \frac{4}{\sqrt{2}} - V_1$$

$$V = 4\sqrt{5} \quad V' = 6\sqrt{2}$$

4.
Sol.


From the figure, it is clear that the distance covered by the plank is l , when cylinder has moved through distance $l/2$.

 5.
Sol.

C
For point A & C, loop BCD shorted

$$\text{Hence } R_{AC} = \frac{r \times 2r}{3r} = \frac{2}{3}r$$

 6.
Sol.

$$V_A - V_B = B(v \sin 30^\circ)l = 4V$$

$$V_B - V_C = -B(v \sin 30^\circ)\frac{l}{2} = -2V$$

 7.
Sol.

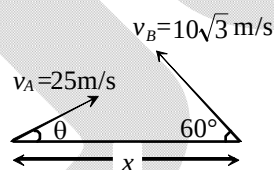
D
For collision,

$$v_A \sin \theta = v_B \sin 60^\circ$$

$$25 \sin \theta = 10\sqrt{3} \times \frac{\sqrt{3}}{2}$$

$$\sin \theta = \frac{3}{5}$$

$$\text{or } \theta = 37^\circ$$


 8.
Sol.

$$A$$

$$\frac{dQ}{dt} = \frac{\pi P r^4}{8\eta L}$$

As capillaries are joined in series, so (dQ/dt) will be same for each capillary.

$$\text{Hence, } \frac{\pi P r^4}{8\eta L} = \frac{\pi P' (r/2)^4}{8\eta (L/2)} = \frac{\pi P'' (r/3)^4}{8\eta (L/3)}$$

So, pressure difference across the ends of 2nd capillary

$$P' = 8P$$

and across the ends of 3rd capillary

$$P'' = 27P$$

 9.
Sol.

C
Time period $\propto n^3$

$$\text{Total time} \propto \frac{1}{n-1}$$

$$\text{No. of revolution} \propto \frac{1}{n^3(n-1)}$$

$$\frac{n_1}{n_2} = \frac{27}{4}$$

10.

D

Sol. Ray inside medium AB is parallel to ray inside medium CD

11.

B

Sol. Gravitational pull depends upon the acceleration due to gravity on that planet.

$$M_m = \frac{1}{81} M_e, g_m = \frac{1}{6} g_e$$

$$g = \frac{GM}{R^2} \Rightarrow \frac{R_e}{R_m} = \left(\frac{M_e}{M_m} \times \frac{g_m}{g_e} \right)^{1/2} = \left(81 \times \frac{1}{6} \right)^{1/2}$$

$$\therefore R_e = \frac{9}{\sqrt{6}} R_m$$

12.

C

$$\text{Sol. } F = -\frac{dU}{dx} = -8 \sin 2x, a = \frac{F}{m} = -8 \sin 2x \quad (m = 1 \text{ kg})$$

for small oscillations $\sin 2x \approx 2x$ i.e. $a = -16x$ since $a \propto -x$

oscillations are simple harmonic in nature

$$\therefore T = \frac{\pi}{2} \text{ s}$$

13.

B

$$\text{Sol. } V_{\max}^2 = 2a_1 \frac{s}{m} t_1 \quad t_1 = \frac{V_{\max}}{a_1}$$

$$V_{\max}^2 = 2a_2 \frac{s}{n} t_2 \quad t_2 = \frac{V_{\max}}{a_2}$$

$$(V) = \frac{s}{t_1 + t_2 + t_3} \quad t_3 = \frac{s \left(1 - \frac{1}{m} - \frac{1}{n} \right)}{V_{\max}}$$

14.

B

Sol. The velocity increases upto $t = 11$ s as acceleration remains positive (although it decreases). $\therefore$ Max. velocity = Area under $v - t$ graph

$$= \frac{1}{2} \times 11 \times 10 = 55 \text{ m/s}$$

15.

B

Sol. From the figure,

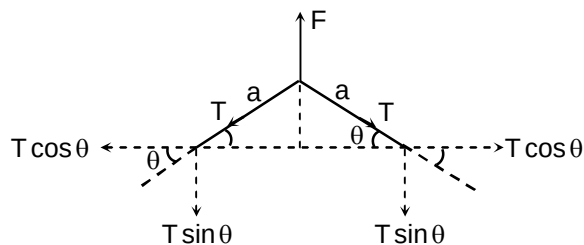
$$F = 2T \sin \theta \quad \dots(i)$$

$$mA = T \cos \theta \quad \dots(ii)$$

Dividing equation (ii) by eqn. (i),

$$\frac{\cos \theta}{2 \sin \theta} = \frac{mA}{F}$$

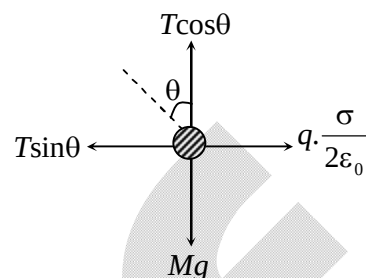
$$A = \frac{F}{2m} \cot \theta$$



$$= \frac{F}{2m} \frac{x}{\sqrt{a^2 - x^2}}$$

16. D

Sol. $T \sin \theta = \frac{q\sigma}{2\epsilon_0}$, $T \cos \theta = Mg$, $\tan \theta \propto \sigma$



17. A

Sol. For first Bohr orbit of hydrogen atom,

$$mvr = \frac{h}{2\pi}, \Rightarrow 2\pi r = \frac{h}{mv}$$

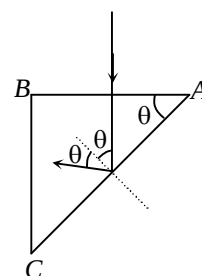
de-Broglie wavelength of electron having same velocity, $\lambda = \frac{h}{mv}$

$$\therefore \frac{2\pi r}{\lambda} = 1$$

18. A

Sol. $\sin \theta > \sin \theta_c = \frac{1}{\mu}$

$$\sin \theta > \frac{1}{\frac{3}{2}}, \sin \theta > \frac{2}{3}$$



19. D

Sol. After collision velocity becomes $v/2$ so radius becomes $a/2$ and hence diameter is a and O will be the centre. So particle strikes at point N .

20. B

Sol. $f = \frac{(2n-1)v}{4l}$

$$l = 25 \text{ cm}, 75 \text{ cm}, 125 \text{ cm}$$

Minimum height of water is 45 cm

SECTION – B

21. 4

Sol. $(Q_i)_B = +2Q$, $Q_f = -2Q$, $\Delta Q = -4Q$
 $\therefore x = 4$

22. 200

Sol. At any time length of each wire $= l - 2vt$

Induced emf $= 4 B v(l - 2vt)$

$$\begin{aligned} \text{Induced current} &= \frac{4Bv(l - 2vt)}{4\lambda(l - 2vt)} = \frac{Bv}{\lambda}, F = B\left(\frac{Bv}{\lambda}\right)(l - 2vt) = \frac{B^2v}{\lambda}(l - 2vt) \\ &= \frac{(2)^2 \times 5}{0.5}(15 - 2 \times 5 \times 1) = 200 \text{ N} \end{aligned}$$

23. 942

Sol. $Z^2 = R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2 \quad \dots (i)$

$$\cos\phi = \frac{R}{Z} = 0.8$$

$$\therefore R = 0.8Z \quad \dots (ii)$$

Putting this value in equation (i)

$$0.64 Z^2 + \left(\omega L - \frac{1}{\omega C}\right)^2 = Z^2$$

$$\therefore \omega L - \frac{1}{\omega C} = 0.6Z \quad \dots (iii)$$

Also given $V_C = 2/5 V_L$

$$I X_C = 2/5 X_L I$$

$$\therefore X_C = 0.4 X_L \quad \dots (iv)$$

$\therefore$ Putting this in equation (iii)

$$\omega L - 0.4\omega L = Z$$

$$\therefore Z = 0.6 \omega L = 0.6 \times 5 \times 314$$

$$Z = 942 \Omega$$

24. 1

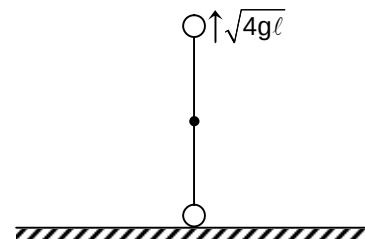
Sol. $m\sqrt{4g\ell} = 2mv$

$$v = \sqrt{g\ell}$$

$$h = \frac{v^2}{2g} = \frac{\ell}{2}$$

Total maximum height of centre of mass from ground

$$= \frac{\ell}{2} + \frac{\ell}{2} = \ell$$



25. 9

 Sol. $L_{\text{vap}} = 540 \text{ cal/g}$
 $L_{\text{fusion}} = 80 \text{ cal/g}$

In steady state,

$$P_1 = K \cdot A \cdot \left\{ \frac{400 - 0}{\lambda x} \right\}$$

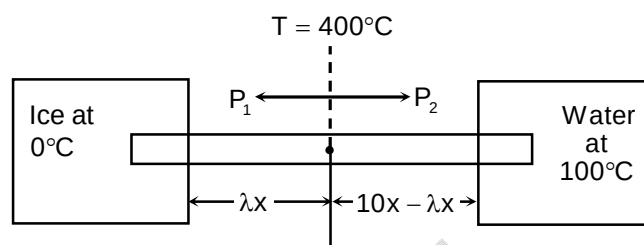
$$\text{And } P_2 = K \cdot A \cdot \left\{ \frac{400 - 100}{10x - \lambda x} \right\}$$

$$\text{Also, } P_1 = \frac{d}{dt} \{ M_{\text{ice}} \cdot L_{\text{fusion}} \}$$

$$\text{and } P_2 = \frac{d}{dt} \{ M_w \cdot L_{\text{vap}} \}$$

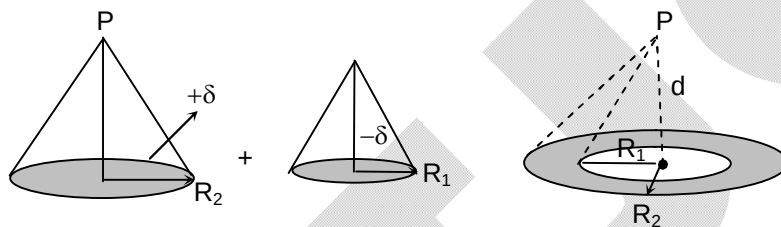
$$\therefore P_1 = L_{\text{fusion}} \cdot \frac{dM_{\text{ice}}}{dt} \text{ and } P_2 = L_{\text{vap}} \cdot \frac{dM_w}{dt}$$

$$\text{Now, } \frac{dM_{\text{ice}}}{dt} = \frac{dM_w}{dt}$$

 Hence, solving, we get $\lambda = 9$.


26. 2

Sol.



$$E = E_1 - E_2 = \frac{\delta}{10\epsilon_0} = 2$$

27. 18

 Sol. We have, $F = (12 - 2t)$

 or, $m \frac{dv}{dt} = (12 - 2t)$, where m is the mass of the body

$$\text{or, } \frac{dv}{dt} = \left(\frac{12 - 2t}{m} \right)$$

$$\int_0^v dv = \frac{1}{m} \int_0^t (12 - 2t) dt$$

$$\text{or, } v = \frac{12t - t^2}{m}$$

$$\text{or, } \frac{dx}{dt} = \frac{(12t - t^2)}{m}$$

$$\int_0^0 dx = \frac{1}{m} \int_0^{t_1} (12t - t^2) dt, \text{ where } t_1 \text{ is required time.}$$

$$\text{or, } 0 = \frac{1}{m} \left[6t_1^2 - \frac{t_1^3}{3} \right]$$

or, $t_1 - 18 = 0$
 $t_1 = 18\text{s}$

28. 30

Sol. When an α -particle is accelerated by 10^4 volt, its kinetic energy will be $K = (2e)(10^4 \text{ V}) = 2 \times 10^4 \text{ eV}$

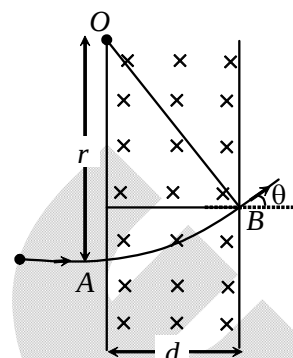
Now the path of a charged particle when it enters a magnetic field at right angles is a circle with radius

$$r = \frac{mv}{qB} = \frac{\sqrt{2mK}}{qB}$$

So here, $r = \frac{(2 \times 6.4 \times 10^{-27} \times 2 \times 10^4 \times 1.6 \times 10^{-19})^{\frac{1}{2}}}{2 \times 1.6 \times 10^{-19} \times 0.1} = 0.2 \text{ m}$

Now as in case of a circle angle between tangents at two points will be equal to the angle between normals at these points. As in a circle tangent is normal to radius at every point, the change in direction of the particle as it passes the field.

$$\theta = \sin^{-1}\left(\frac{d}{r}\right) = \sin^{-1}\left(\frac{0.1}{0.2}\right) = 30^\circ$$



29. 56

Sol. Net force upward = weight of liquid in capillary.

$$\left[\frac{2\pi d_2}{2} T + \frac{2\pi d_1}{2} T \right] = \pi \left(\frac{d_2^2}{4} - \frac{d_1^2}{4} \right) \rho g h$$

$$h = 56 \text{ mm}$$

30. 4

Sol. $\tan \frac{\pi}{6} = \frac{X_L}{R} = \frac{1}{\sqrt{3}}$

$$R = 3\Omega$$

$$\tan \theta = \frac{X_C}{R} = 1\Omega$$

$$\phi = \frac{\pi}{4}$$

Chemistry

PART – B

SECTION – A

31. B
Sol. P = C₂ epimer
32. B
Sol. $\sqrt{n(n+2)}$ = Magnetic moment
Both are d⁹ with 1 unpaired e⁻
33. A
Sol. Geometry = cis/trans
Linkage = SCN
34. B
Sol. Factual
35. D
Sol. $\text{MnO}_4^- + \text{e}^- \rightarrow \text{MnO}_4^{2-}$ –alkaline medium
 $\text{MnO}_4^- + \text{e}^- \rightarrow \text{MnO}_2$ – neutral medium
36. C
Sol. Self-dipeptide of glycine gives 1 – optically inactive.
Self-dipeptide of L valine gives 1 – optically active
Cross dipeptide of glycine + L valine gives 2 – both optically active
37. D
Sol. $\Delta G^\circ = -nFE^\circ_{\text{cell}}$
38. C
Sol.
 - (2, 4-DNP) test given by aldehyde 1 ketones
 - Tollen test
 - Canizaro
 - Must be at 1, 2 position
39. A
Sol. $\text{pH} = \text{pK}_a + \frac{\log[\text{CH}_3\text{COO}^-]}{\text{CH}_3\text{COOH}}$
40. A
Sol. $\int_{T_1}^{T_2} dH = \int_{T_1}^{T_2} C_p dT$
C_p is temperature dependent
41. C
Sol.
 - CFT
 - Magnetic moment = $\sqrt{n(n+2)}$ B.M
42. A
Sol. Factual

43. B
Sol. $K_a = \text{III} > \text{IV} > \text{I} > \text{III}$
 $pK_a = \text{II} > \text{I} > \text{IV} > \text{III}$

44. B
Sol. $\frac{-dN}{dt} = \lambda N$

First order kinetic, $t_{1/2} = \frac{0.693}{\lambda}$

45. B
Sol. $i = 2$
 $\pi = iCRT$

46. C
Sol. Properties of electromagnetic Waal's theory

47. D
Sol. VSEPR

48. A
Sol.
$$\begin{bmatrix} \Delta G^\circ = -nFE_{\text{cell}}^\circ \\ \Delta S^\circ = nF \left(\frac{\delta E_{\text{cell}}^\circ}{\delta T} \right) \\ \Delta H = \Delta G^\circ + T\Delta S^\circ \end{bmatrix}$$

49. B
Sol. $[R] = \frac{K_2}{K_1 + K_2} [P] (1 - e^{-(k_1 + k_2)t})$

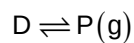
50. B
Sol. % ionic character = $\frac{\text{observed dipole moment}}{\text{calculated dipole momet}} \times 100$

SECTION – B

51. 5
Sol. $7\alpha + 4\beta = x$
 $x = 11$

52. 2
Sol. $\frac{[B]}{[C]} = \frac{k_1}{k_2} = \frac{2k_2}{k_2} = 2$

53. 5
Sol. $K_{\text{final}} P(g) \rightleftharpoons D(g) = K_{C_1} \times K_{C_2} \times K_{C_3}$
 $= 10 \times 2 \times 0.01 = \frac{20}{100} \times \frac{2}{10}$
 $= 1/5$



$$K_f^1 = \frac{1}{K_{\text{final}}} = \frac{1}{1/5} = 5$$

54. 3

 Sol. CFT, H₂O weak field ligand

55. 3

$$\text{Sol. } K_a = \frac{C\alpha^2}{1-\alpha}$$

$$\sqrt{K_a/c} = \alpha$$

56. 4

$$\text{Sol. } E_n = -\frac{13.5}{n^2} Z^2$$

57. 3

Sol. Paramagnetic species check by MOT

58. 2

$$\text{Sol. } t_{1/2} \propto \frac{1}{A_0^{n-1}}$$

59. 2

Sol. Rate becomes double.

60. 5

Sol. Mole concept

Mathematics**PART – C****SECTION – A**

61. B

Sol. $\sin^{-1} \sqrt{\frac{2-\sqrt{3}}{4}} = \frac{\pi}{12}$
 $\cot\left(\frac{\pi}{2}\right) = 0$

62. D

Sol. $\frac{\tan 3\theta}{\tan \theta} = 4 \Rightarrow \tan^2 \theta = \frac{1}{11}$
 Now, $\frac{\sin 3\theta}{\sin \theta} = 3 - 4 \sin^2 \theta = 3 - 4\left(\frac{1}{1 + \cot^2 \theta}\right) = \frac{8}{3}$

63. B

Sol. Elements of set A are in $4n + 2$ from, $n \in \mathbb{N}$
 $360 = 2^3 \times 3^2 \times 5^1$
 Sum of common elements
 $= 2[(3^0 + 3^1 + 3^2)(5^0 + 5^1) - 1] = 154$

64. A

Sol. $f(x) = \sqrt{\ln(2\lambda \cos x + 5)}$
 Nece. Condition $2\lambda \cos x + 5 \geq 1$
 $\Rightarrow \lambda \cos x \geq -2 \forall x \in \mathbb{R}, \lambda = 0, \pm 1, \pm 2$

65. B

Sol. The given equation can be written as

$$(x dy + y dx) + \left\{ \ln x dy + \frac{y}{x} dx \right\} + (\sin y dx + x \cos y dy) = 0$$

$$\Rightarrow d(xy) + d(y \ln x) + d(x \sin y) = 0$$

On integrating, we get
 $xy + y \ln x + x \sin y = c$
 Where c is arbitrary constant

66. A

Sol. $\therefore HG : GO = 2 : 1$

$$\therefore G = \left(1, \frac{8}{9}\right)$$

Hence, mid-point of BC = $\left(1, -\frac{11}{3}\right)$

67. A

Sol. $4a = 2p$

$$\therefore a = \frac{p}{2}$$

Direction is $x + a = 0$ or $x + \frac{p}{2} = 0$

$$\left(x - \frac{p}{2}\right)^2 + y^2 = r^2 \text{ touches } x + \frac{p}{2} = 0$$

$\therefore$ condition of tangency given $r = p$

$$\therefore \therefore \left(x - \frac{p}{2}\right)^2 + y^2 = p^2 \text{ and } y^2 = 2px$$

Putting for y^2 , we get $4x^2 + 4px - 3p^2 = 0$

$\therefore x = p/2, -3p/2$ (rejected)

As x cannot be -ive because in that case $y^2 = 2px = -ive$

When $x = p/2$, then $y^2 = p^2 \therefore y = p, -p$

68. B

Sol. Range of $a = \left(-\frac{1}{2}, \frac{1}{2}\right)$ since $y - y^2 = a^2$

$$\Rightarrow \Delta > 0 \Rightarrow a \in \left(-\frac{1}{2}, \frac{1}{2}\right)$$

$$GE = 1$$

69. B

$$\text{Sol. } \vec{\alpha} \cdot \vec{\beta} = ab + bc + ca = \sqrt{a^2 + b^2 + c^2} \sqrt{b^2 + c^2 + a^2} \cos \theta \Rightarrow \cos \theta = \frac{ab + bc + ca}{(a^2 + b^2 + c^2)}$$

$$\text{Now } (a - b)^2 + (b - c)^2 + (c - a)^2 \geq 0$$

$$a^2 + b^2 + c^2 \geq ab + bc + ca \Rightarrow \frac{ab + bc + ca}{a^2 + b^2 + c^2} \leq 1$$

$$\text{Also } (a + b + c)^2 = a^2 + b^2 + c^2 + 2(ab + bc + ca) \geq 0$$

$$\Rightarrow \frac{ab + bc + ca}{a^2 + b^2 + c^2} \geq -1/2$$

$$\Rightarrow -\frac{1}{2} \leq \cos \theta \leq 1 \Rightarrow \theta \in [0, 2\pi/3]$$

70. D

$$\text{Sol. } \begin{vmatrix} (a^x + a^{-x})^2 & (a^x - a^{-x})^2 & 1 \\ (b^y + b^{-y})^2 & (b^y - b^{-y})^2 & 1 \\ (c^z + c^{-z})^2 & (c^z - c^{-z})^2 & 1 \end{vmatrix}$$

Operating $C_1 \rightarrow C_1 - C_2$

$$\begin{vmatrix} 4a^x a^{-x} & (a^x - a^{-x})^2 & 1 \\ 4b^y b^{-y} & (b^y - b^{-y})^2 & 1 \\ 4c^z c^{-z} & (c^z - c^{-z})^2 & 1 \end{vmatrix} = 4 \begin{vmatrix} 1 & (a^x - a^{-x})^2 & 1 \\ 1 & (b^y - b^{-y})^2 & 1 \\ 1 & (c^z - c^{-z})^2 & 1 \end{vmatrix} = 0$$

71. C

Sol. $0.7 = P(A \cup B) = P(A) + P(B) - P(A \cap B) = P(A) + P(B) - P(A) \cdot P(B)$
 $= 0.2 + P(B) - 0.2P(B)$
 $\Rightarrow P(B) = \frac{0.5}{0.8} = \frac{5}{8}$
 $\Rightarrow P(B') = 3/8.$

72. B

Sol. $\sum_{r=1}^{1024} [\log_2 r] = \sum_{r=1}^{2^{10}} [\log_2 r]$
 $= \sum_{r=2}^{2^2-1} [\log_2 r] + \sum_{r=2^2}^{2^3-1} [\log_2 r] + \sum_{r=2^3}^{2^4-1} [\log_2 r] + \dots + \sum_{r=2^9}^{2^{10}-1} [\log_2 r] + \log_2 2^{10}.$
 $= 2.1 + 2^2 \cdot 2 + 2^3 \cdot 3 + 2^4 \cdot 4 + \dots + 2^9 \cdot 9 + 10$
 $= \sum_{r=1}^9 2^r \cdot r + 10 = 8204.$

73. A

Sol. Total number of numbers will be equal to the sum of numbers of all possible 1-digit, 2-digit, 3-digit, 4-digit and 5-digit numbers.
 $\Rightarrow$ Total number of numbers $= 3 + 3^2 + 3^3 + 3^4 + 3^5$
 $= \frac{3(3^5 - 1)}{2} + 3^4 = \frac{3^5 + 2 \cdot 3^4 - 3}{2}.$

74. A

Sol. Given expression $= i \left[\log \left(\frac{x-i}{x+i} \right) \right] - \pi + 2 \tan^{-1} x = k$ (say)

$$\log \left(\frac{x+i}{x-i} \right) = (k + \pi - 2 \tan^{-1} x)i$$

or $\frac{x+i}{x-i} = e^{i\theta}$ where $\theta = k + \pi - 2 \tan^{-1} x$

$$\Rightarrow (x+i) = (x \cos \theta + \sin \theta) + i(x \sin \theta - \cos \theta)$$

$$x = x \cos \theta + \sin \theta \text{ and } 1 = x \sin \theta - \cos \theta$$

$$\Rightarrow x = \cot(\theta/2) \Rightarrow \theta = 2 \cot^{-1} x$$

$$\text{or } k + \pi - 2 \tan^{-1} x = 2 \cot^{-1} x$$

$$\text{or } k + \pi = 2 [\tan^{-1} x + \cot^{-1} x] = 2(\pi/2)$$

$$\text{or } k + \pi = \pi$$

$$k = 0.$$

75. C

Sol. $3x^2 - 2(a-b+b-c+c-d)x + (a-b)^2 + (b-c)^2 + (c-d)^2 = 0$
 $(x-a+b)^2 + (x-b+c)^2 + (x-c+d)^2 = 0$
 $\Rightarrow x = a-b = b-c = c-d$
 $\therefore a, b, c, d$ are in A.P.

76. B

Sol. We have $a \sin x + 2b \cos C \cos x = \alpha$

The maximum value of $a \sin x + 2b \cos C \cos x$ is $\sqrt{a^2 + 4b^2 \cos^2 C}$ which increases as $|\cos C|$ increases. So the minimum value of $|\cos C|$ will correspond to the situation when $\sqrt{a^2 + 4b^2 \cos^2 C}$ is just equal to α , so as to give a real solution of the equation.

$$\therefore \sqrt{a^2 + 4b^2 \cos^2 C} = \alpha$$

$$\cos^2 C = \frac{\alpha^2 - a^2}{4b^2}; |\cos C| = \sqrt{\frac{\alpha^2 - a^2}{4b^2}}$$

77. B

Sol. $\frac{y dx - x dy}{y^2} + x dx = 0$

$$d\left(\frac{x}{y}\right) + d\left(\frac{x^2}{2}\right) = 0$$

Integrating both sides

$$\frac{x}{y} + \frac{x^2}{2} = \lambda$$

78. D

Sol. $|\sin x| \geq \sin^2 x$

$$|\cos x| \geq \cos^2 x$$

$$|\sin x| + |\cos x| \geq 1$$

$$1 \leq |\sin x| + |\cos x| \leq \sqrt{2}$$

$$\text{Thus } [|\sin x| + |\cos x|] = 1$$

$$\int_0^{2\pi} [|\sin x| + |\cos x|] dx = \int_0^{2\pi} 1 dx = 2\pi$$

79. B

Sol. Let $f(x) = \frac{ax^4}{4} + \frac{bx^3}{3} + \frac{cx^2}{2} + dx$, which is continuous and differentiable.

$$f(0) = 0, f(-1) = \frac{a}{4} - \frac{b}{3} + \frac{c}{2} - d = \frac{1}{4}(a + 2c) - \frac{1}{3}(b + 3d) = 0$$

so, according to Rolle's theorem, there exist atleast one root of $f'(x) = 0$ in $(-1, 0)$.

80. D

Sol. $\bar{x} = 8 \Rightarrow \sum x_i = 7 \times 8 = 56$

$$42 + a + b = 56$$

$$a + b = 14$$

$$\sigma^2 = 16$$

$$\frac{\sum x_i^2}{n} - \bar{x}^2 = 16$$

$$\frac{4 + 16 + 100 + 144 + 196 + a^2 + b^2}{7} - 64 = 16$$

$$a^2 + b^2 = 560 - (460) = 100$$

$$\frac{(a+b)^2 - (a^2 - b^2)}{2} = ab$$

$$\frac{196 - 100}{2} = 48 = ab$$

SECTION - B

81. 3

Sol. $f'(x) = \frac{1}{(f(x))^2} \Rightarrow f'(x)f(x)^2 = 1 \Rightarrow \frac{(f(x))^3}{3} = x + c$

Put $x = 2$, we get $\frac{6}{3} = 2 + c \Rightarrow c = 0$

$\therefore f(x) = (3x)^{\frac{1}{3}} \Rightarrow f(9) = 3$

82. 40

Sol. $2ac = 6 \quad e = \frac{3}{5} \Rightarrow a = 5$

Area of quadrilateral

$$= 2ab$$

$$= 2a^2 \sqrt{1 - e^2}$$

$$= 2 \times 25 \sqrt{1 - \frac{9}{25}}$$

$$= 2 \times 25 \times \frac{4}{5} = 40$$

83. 8

Sol. $(MN)^2 = 9MN$

Also $(NM)^3 = N(MN)^2M = N(9MN)M = 9(NM)^2$

$$\Rightarrow NM = 9$$

84. 3

Sol. $n! + (n+1)! + (n+2)! = n! \{1 + n + 1 + (n+2)(n+1)\}$

$$= n!(n+2)^2$$

$$\Rightarrow \text{Either 7 divides } n+2 \text{ or 49 divides } n!$$

$$\Rightarrow n = 5, 12, 14.$$

85. 0

Sol. All the terms in the expansion of $\left(x + \frac{1}{x}\right)^{11}$ will have odd powers of x .

$$\text{So required sum} = 0.$$

86. 3

Sol. The function is discontinuous at $x = 0, \frac{1}{2}$ and 2.

87.

1

Sol.

Given equation of the curve is

$$x + y - \ln(x + y) = 2x + 5$$

$$\Rightarrow 1 + \frac{dy}{dx} - \frac{1}{(x + y)} \left(1 + \frac{dy}{dx} \right) = 2$$

$$\Rightarrow \frac{dy}{dx} \left(1 - \frac{1}{x + y} \right) = 1 + \frac{1}{x + y}$$

$$\Rightarrow \frac{dy}{dx} = \frac{x + y + 1}{x + y - 1} \Big|_{(\alpha, \beta)} = \infty \Rightarrow \alpha + \beta = 1.$$

88.

3

Sol.

$$x^2 + y^2 + z^2 = 2x + 2y + 2z - 3$$

$$\Rightarrow x^2 - 2x + 1 + y^2 - 2y + 1 + z^2 - 2z + 1 = 0$$

$$\Rightarrow (x - 1)^2 + (y - 1)^2 + (z - 1)^2 = 0$$

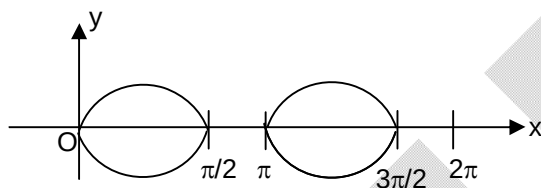
$$\Rightarrow x = 1, y = 1, z = 1$$

$$x + y + z = 3.$$

89.

4

Sol.



$$\begin{aligned} \text{Required area} &= 4 \int_0^{\pi/2} (\sin 2x) dx \\ &= 4 \text{sq. unit} \end{aligned}$$

90.

8

Sol.

Any plane through the second line is $(3x - y - 2z + 4) + \lambda(2x + y + z + 1) = 0$

Or $(3 + 2\lambda)x + (\lambda - 1)y + (\lambda - 2)z + 4 + \lambda = 0$.

If this plane is parallel to the first line then its normal must be at right angle to first line.

$$\Rightarrow (3 + 2\lambda)2 + (\lambda - 1)4 + (\lambda - 2) = 0 \Rightarrow \lambda = 0$$

$\Rightarrow$ equation of plane through the second line and parallel to the first line is

$$3x - y - 2z + 4 = 0 \quad \dots\dots(1)$$

Now the required shortest distance s = perpendicular distance of a point $(1, 3, -2)$ on the first line

$$\text{to the plane } 3x - y - 2z + 4 = 0 = \frac{8}{\sqrt{14}} \Rightarrow s\sqrt{14} = 8.$$